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Reactive power boosting strategy for hydrogenerators: framework and evaluation using Nordic machine data

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Modern power systems face operational challenges due to the integration of variable renewable energy, necessitating enhanced reactive power support from synchronous generators. In this study, we propose a comprehensive reactive power boosting strategy (Q-strategy) framework, addressing technical and economic challenges via four pillars: (1) Q-boost defines reactive power limits for continuous/dynamic states by leveraging thermal margins under insulation thresholds (155 °C); (2) Q-model, a capacity twin model (CTM) combining a lumped-parameter thermal network and saturated field current models, enables online hot spot temperature management without finite element analysis; (3) Q-energy optimizes reactive energy delivery by integrating thermal endurance models with aging and start–stop cycles; and (4) Q-value formulates costs for reactive power services, accounting for losses beyond grid codes and lifetime depreciation. The results demonstrate a 57% increase in continuous reactive power capacity, with smaller generators incurring higher costs (up to 1.33 $/Mvarh) than larger units. The proposed framework exploits hydrogenerators' slow thermal time constants by using authentic historical machine parameters and operational statistics from Nordic installations. To isolate the machine's direct electro-thermal boundaries and localized voltage-support capabilities without confounding multi-bus network variables, system-level validation is conducted on a single-machine, two-bus power system topology subjected to extreme contingency events and localized load profiles. An online temperature controller ensures safe operation under extreme conditions while aligning with real-world load patterns. The Q-strategy offers a cost-effective alternative to FACTS devices, balancing grid resilience with economic feasibility in high-renewable grids.

KEYWORDS
lumped-parameter thermal network, 3D heat flux FEM calculations, rotor copper hotspots dynamics, stator copper hotspots dynamics, enhanced voltage stability, reactive power boosting

1 Introduction

The energy transition is driving power systems toward greater reliance on clean energy and electrification, particularly through the deployment of renewable energy

sources such as solar and wind. It is widely expected that by 2050, 75% of electricity generation will come from inverter-based variable renewable energy (VRE), and electricity will account for 52% of total energy consumption (Barani et al., 2026). Although this shift significantly reduces net carbon emissions, it alters the physical architecture of transmission grids, necessitating a comprehensive re-evaluation of structural voltage-support mechanisms.

Modern power systems face acute operational stress from electronic loads, HVDC links, and VRE, which can trigger cascade outages, particularly affecting voltage stability following major disturbances (Adegoke et al., 2024). The fundamental operational problem stems from a stark physical disconnect in reactive current capabilities during grid faults. Traditional power systems relied on heavy iron-and-copper synchronous assets whose large mass naturally withstood large instantaneous overcurrent injections, thereby supporting weak voltage profiles by virtue of their thermal capacity. In contrast, the power-electronic converter interfaces used in VRE and HVDC links have virtually zero thermal overload capacity; to protect fragile semiconductor switches from instantaneous destruction, their internal control loops strictly clip the output current within microseconds during a fault. This structural shift deprives modern networks of the immediate, dynamic, reactive current overshoots that would otherwise mitigate rapid localized voltage degradation and accelerate wide-area voltage collapse (Liao et al., 2026).

Maintaining grid security under these current-constrained conditions requires distinct control mechanisms operating across different timelines. Long-term voltage stability depends on operator control actions and slower-acting equipment, such as generator current limiters. In contrast, short-term voltage stability relies on fast-acting grid infrastructure, including protective devices, FACTS, and synchronous condensers (Dahat and Dhabale, 2024; Nair et al., 2024; Hatziargyriou et al., 2021). Synchronous condensers are typically deployed near HVDC links to improve localized system strength and provide missing short-circuit current capacity (Duan et al., 2022; Tan et al., 2022), an objective that can be advanced by integrating specialized machine-level current control and megavolt-ampere (MVA)-field limiters directly into the excitation loop to enable rapid localized voltage response (Ojo and Shafiul Alam, 2026). The severity of these structural deficits is well documented; for instance, a study using a modified Nordic test system with a 35% share of inverter-based generation shows an increased risk of blackouts due to voltage instability (Heusinger et al., 2021). Moreover, there are currently no clear plans to introduce structural inertia or dynamic reactive reserves as balancing services in the Nordic system (Nordström et al., 2023), reflecting a broader global challenge where the rapid displacement of traditional plants by non-synchronous assets severely degrades baseline grid strength, rotational reserves, and multi-timeline voltage resilience (Dulal et al., 2026).

As VRE penetration increases, conventional synchronous generators are displaced from routine energy production and repurposed as synchronous condensers to maintain relevance and enhance wide-area stability (Halley et al., 2022; Gang et al., 2022; Bichler et al., 2022). Although this repurposing strategy addresses immediate network needs such as baseline inertia and basic reactive power support, existing hydrogenerators offer a unique, highly localized opportunity to provide dynamically enhanced reactive power support while still providing active power (Oyvang and Noland, 2022). However, running these heavy assets under modern, highly variable non-nominal regimes increases thermal cycling and structural stress, necessitating physical refurbishment or rotor pole re-optimization to balance asset lifetime degradation against operational performance benefits (Karekezi et al., 2025). This boosting opportunity stems directly from an unexploited margin in historical manufacturing design: hydrogenerators traditionally operate at conservative Class B insulation limits (130 °C), even though they are frequently manufactured with high-tier Class F insulation materials rated for continuous operation at 155 °C (Pyrhönen et al., 2013). Accounting for a standard 10 K internal hot spot margin, this difference provides an unexploited 25 K physical temperature margin for reactive power boosting, serving as a thermal buffer for the grid during critical voltage-stability contingencies and system optimization via optimal reactive power dispatch (ORPD).

Although the domain of reactive power management has been investigated across separate technical and economic subfields, a critical gap persists in linking active component-level boosting capabilities directly with macro-level market cost-formation architectures. Recent state-of-the-art studies have separate pockets of innovation. On the control infrastructure side, Banda et al. (2025) introduced a novel control strategy to achieve reactive power boosting from single-stage BESS-STATCOM systems, showing that isolating the battery pack can maximize the dc-link capacitor's voltage capability, thereby overcoming low-state-of-charge performance bottlenecks. On the economic and regulatory side, Chiandone et al. (2025) advanced the valuation of dynamic reactive services by proposing a cost allocation framework that accounts for variations in synchronous generator efficiency during under- or overexcitation phases to establish a fair merit order for secondary voltage regulation. However, these streams of research remain decoupled. Hardware-centric power electronic control models are not applicable to heavy iron-and-copper hydro-turbines, and generalized generator efficiency pricing models typically omit the localized insulation thermal degradation, copper loss escalations, or transient hotspot accumulation that physical over-excitation boosting imposes on machine winding profiles.

To bridge this gap, we propose a comprehensive, self-contained mathematical and economic framework for introducing new dynamic reactive power services. This framework directly connects high-fidelity lumped-parameter thermal networks for hotspots with macro-level asset-valuation models, providing clear economic incentives for power plant owners to optimize their reactive capabilities in real time. The ultimate objective is to deliver extended voltage and inertial-like support during extreme grid scenarios through a tightly integrated four-pillar architecture:

- Q-boost: Defines the maximum absolute reactive power output that a machine can physically provide by safely operating beyond conservative operational boundaries.
- Q-model: Establishes a real-time mathematical framework (capacity twin model, CTM) for monitoring and controlling internal hot spot temperatures without heavy computational overhead.

- **Q-energy:** Integrates reactive power delivery over time, defining explicit, safe operational time windows for extended boosting without endangering the asset.
- Q-value: Introduces a comprehensive, asset-degradation- and physical-loss-aware cost formation mechanism that ensures sustainable and fair market compensation for hydrogenerator assets.

2 Problem formulation

The untapped potential of hydrogenerators must be adopted to enhance grid optimization and voltage stability margins without incurring extensive capital expenditure on additional FACTS devices, particularly under extreme conditions imposed by a high share of VREs. In this study, we recognize the over-excitation limiter (OEL) as a critical limitation to the hydrogenerator's untapped potential of Q-boost operation. It proposes a Q-strategy framework to leverage the thermal capabilities of hydrogenerators beyond the standard operation schemes. These thermal capabilities are constrained by the insulation class temperature threshold, aging, and the annual increases in start–stop cycles. The hot spot temperatures require accurate and online predictions to prevent failures and minimize wear and tear during Q-boost operation. Furthermore, the lack of integration between reactive power costs, additional losses incurred by hydrogenerators, and lifetime insulation costs hinders the economic incentives for power producers to provide new reactive power services.

2.1 Voltage stability margin enhancement

Reactive power availability is a critical factor that limits the voltage stability margin, especially when restoring the voltage to its optimal level after power system disturbances. If the system's reactive power reserve is insufficient, it can lead to a voltage collapse and a system blackout. Hence, generators with Q-boost capabilities already in the grid are a practical and cost-effective solution for providing the system's reactive power reserves and enhancing voltage stability during disturbances.

The reactive power boundaries of the hydrogenerator are established under nominal conditions for insulation class F (155 °C) to operate at a maximum hot spot temperature of 130 °C, known as the classical Philosophy of Design (POD). Building on the classical hydrogenerator POD with the actual temperature threshold of insulation class F, Equation 1 introduces an approximated Q-boost calculation based on the relation between heat flux and current density (Wang and Ruan, 2022). The variables are primarily substituted to show the field current density's dependency on temperature. At the same time, the remainder, namely, electrical conductivity, pole coil width, the cross-sectional area of the conductor, and thermal resistance, are considered constant values. To facilitate the analysis, the concept of temperature per unit—defined as the ratio of maximum to nominal operating temperature—is introduced, enabling the evaluation of field current per unit at various temperatures. The results indicate that the field current can be increased by 9.19%, allowing for Q-boost without exceeding the thermal limits of insulation class F.

$$K = \frac{1}{\sqrt{b_f \lambda_e R_{hr} A_{hr}}} = \frac{J_f}{\sqrt{T_{hr}}} = \frac{J_f^{nom}}{\sqrt{T_{hr}^{nom}}} \Rightarrow J_f^{pu} = I_f^{pu} = \sqrt{T_{max.}^{pu}}. \quad (1)$$

An example of voltage stability margin enhancement is presented in Figure 1 to demonstrate the impact of Q-boost on system performance, where the field current is obtained using Equation 2 for G1 in Table 1. In this scenario, the active power is set to $P = 0.7767$ pu = 80 MW, resulting in a reactive power of $Q_{boost} = 0.6453$ pu = 66.47 Mvar. Consequently, the voltage stability margin increases by 21%, attributed to an additional 50% reactive power after losing a transmission line, as depicted in Figure 2. In red, the PV curve for the normal operation is shown. The loss of one line without Q-boost is exhibited in green. Finally, the increased voltage stability margin achieved using Q-boost is depicted in blue. The generator terminal voltage without Q-boost is 1.025 pu, and that with Q-boost is 1.05 pu. The total line reactance is $X_L = 0.25$ pu.

$$\begin{aligned}
&I_a = \frac{\sqrt{P^2 + Q^2}}{V_a}, \quad \phi = \arctan(Q, P), \\
&\delta = \arctan\left(X_q I_a \cos(\phi) - R_s I_a \sin(\phi), V_a + R_s I_a \cos(\phi) + X_q I_a \sin(\phi)\right), \\
&E_g = V_a \cos(\delta) + R_s I_a \cos(\delta + \phi) + X_d I_a \sin(\delta + \phi), \\
&\theta = \arctan\left(\frac{X_p I_a \cos(\phi) - R_s I_a \sin(\phi)}{V_a + R_s I_a \cos(\phi) + X_p I_a \sin(\phi)}\right), \\
&E_p = V_a \cos(\theta) + R_s I_a \cos(\phi + \theta) + X_p I_a \sin(\phi + \theta), \\
&I_f = \frac{\left(E_g - E_p\right) + 1.0308 \cdot \left(E_p + 0.16 \cdot E_p^7\right)}{I_f^{ref}}
\end{aligned} \quad (2)$$

2.2 Capacity twin model

A CTM is proposed to achieve online thermal management using a fast H-LPTN, a per-unit loss model, and an electrical model with saturation. The H-LPTN is developed with parameter estimation using actual data for average temperature and finite element analysis (FEA) calculations for hotspots, and subsequently, cross-validated for ultra- and high-loss scenarios; the per-unit loss model calculates the field winding, stator armature, and iron core losses of hydrogenerators (Bortoni et al., 2021), and the saturated field current model (Karekezi et al., 2022) is developed using a polynomial of the form $I_f = a_1(E_p + a_2 \cdot E_p^n)$, where n is usually 7 or 9. These three models complete the thermo-electrical modeling for the CTM. In this study, $a_1 = 1.0308$ and $a_2 = 0.16$. It should be noted that the critical component for thermal limits during reactive power boosting is the rotor hotspot for thermal dynamics slower than 10 min.

The H-LPTN implemented in this study is shown in Figure 3, consisting of six ordinary differential equations (ODEs) and the remaining algebraic relations for the heat exchanger dynamics. This low-order H-LPTN is constructed as a double-layer LPTN that connects the hot spot LPTNs to their nearest heat sources in the system LPTN, i.e., the average temperatures of the hydrogenerator.

The electrical model for developing, calculating, and validating the capability diagram, Q-boost limits, and the Q-value costs uses the field saturation model developed by Karekezi et al. (2022). On

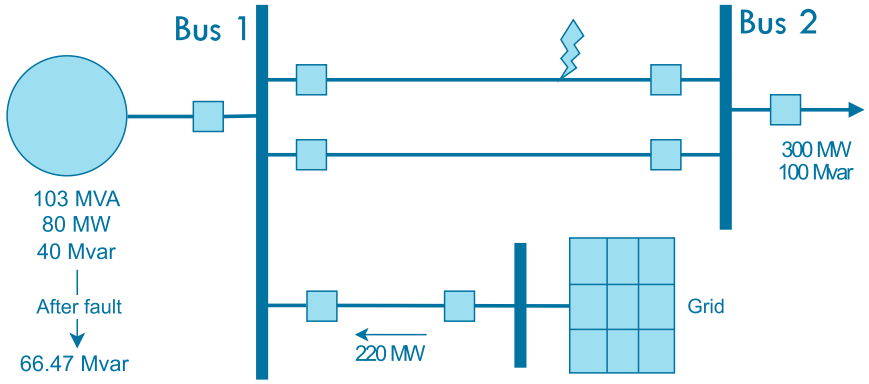


FIGURE 1
Two-bus power system. Reactive power boosting (Q-boost) after a fault for the increased voltage stability margin.

TABLE 1 Generator parameters.

Parameter	Unit	G1	G2	G3	G4
$\cos\phi$	[-]	0.9	0.86	0.86	0.86
X_d	[p.u.]	1.059	1.2	1.2	1.41
X_q	[p.u.]	0.676	0.66	0.63	0.75
X_p	[p.u.]	0.141	0.11	0.21	0.13
R_s	[p.u.]	0.0016	0.0014	0.0011	0.0005
I_f^{ref}	[p.u.]	2.028	2.1709	2.2889	2.3735
ΔP_c	[kW]	211.92	150	320	42.77
ΔP_s	[kW]	276.62	246	387	56.9
ΔP_{fbr}	[kW]	175.78	130	235	33.68
ΔP_{ex}	[kW]	15.88	9	16	5.05
ΔP_{v80}	[kW]	423.32	305.6	516.8	105.28
R_δ	[K/MW]	20.2171	30.4842	21.3249	13.702
R_w	[K/MW]	2.8998	7.7326	3.5259	13.132
S	[MVA]	103	60	120	12
$m_{cu,s}$	[kg]	6827	5150	9700	1084
$m_{cu,r}$	[kg]	9260	7200	13000	1759

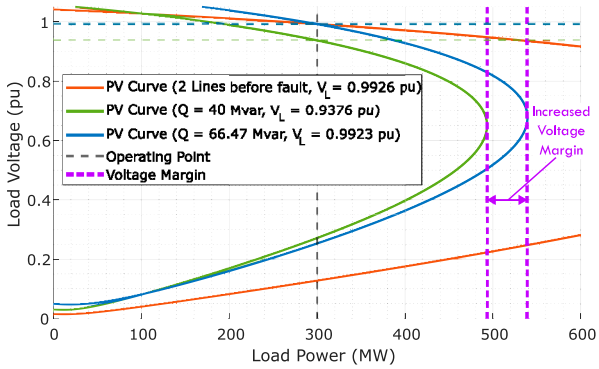


FIGURE 2
PV curves. Voltage stability margin.

the other hand, the open-circuit characteristic of G1 is directly incorporated into the machine block for the case studies performed in Simulink, as described in Section 4.

2.3 Scaled capacity twin model through thermal scaling

To achieve broader applicability across multiple hydrogenerators using the CTM without relying on computationally expensive FEA, thermal parameter scaling is implemented to create a scaled capacity twin model (s-CTM) that approximates the hot spot LPTNs for air-cooled hydrogenerators, i.e., synchronous generators with similar POD. The parameters to scale are the thermal resistances R_{th}, the thermal capacitances $C_{p_{th}}$, and, in consequence, the thermal time constant τ_{th}. The scaling procedure is as follows:

$$R_{th}^{new} = R_{th}^{old} \cdot \frac{\Delta P^{old}}{\Delta P^{new}} \;\wedge\; C_{p_{th}}^{new} = C_{p_{th}}^{old} \cdot \frac{m_{cu}^{new}}{m_{cu}^{old}} \;\wedge\; \tau_{th}^{new} = \tau_{th}^{old} \cdot \frac{\Delta P^{old}}{\Delta P^{new}} \cdot \frac{m_{cu}^{new}}{m_{cu}^{old}}. \tag{3}$$

The primary assumption for thermal scaling is the classical POD of a temperature increase of $\Delta T = 80$ K, as reported by Pyrhönen et al., 2013, which is used to determine the thermal resistance ($R_{th} = \frac{\Delta T}{\Delta P}$). The thermal capacitance ($C_p = c_p \cdot m_{cu}$) assumption is straightforward: the material is the same; only the mass changes. Therefore, all the thermal parameters, including those of the hot spots and end-winding, can be scaled accordingly using the masses and losses of the actual generator.

2.4 Quadratic thermal model

The temperature-dependent reactive power limits can be approximated using Equation 6—referred to as the quadratic

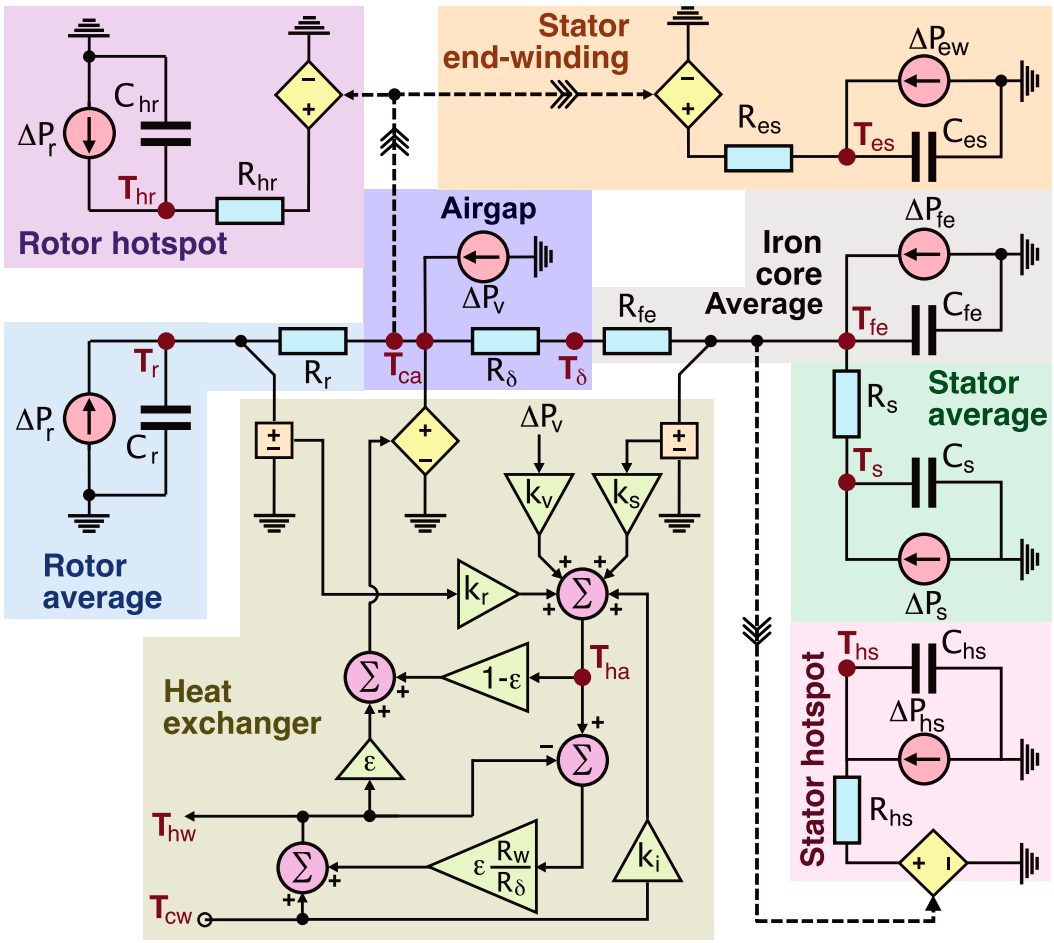


FIGURE 3
Lumped-parameter thermal network for hot spots.

model—which is obtained by combining the rotor hot spot at thermal equilibrium (Equation 4, shown in Figure 3), the rotor loss estimation (Equation 5; Bortoni et al., 2021), and the thermal scaling rule (Equation 3).

$$\frac{dT_{rh}}{dt} = \frac{1}{\tau_{hr}}\left[\Delta P_r \cdot R_{hr} - \left(T_{hr} - T_{ca}\right)\right] \rightarrow \Delta P_r = \frac{T_{hr}^{\infty} - T_{ca}^{\infty}}{R_{hr}}, \quad (4)$$

$$\Delta P_r = \Delta P_f \cdot I_f^2 + \Delta P_{br} \cdot I_f^2 + \Delta P_{ex} \cdot I_f, \quad (5)$$

$$I_f = I_f\left(T_{hr}^{\infty}, T_{cw}, \Delta P_f, \Delta P_{ex}, \Delta P_r\right) \Leftrightarrow \left(\Delta P_f + \Delta P_{br}\right) \cdot I_f^2 + \Delta P_{ex} \cdot I_f + c = 0$$
$$\text{where } c = -\frac{\Delta P_r}{78.6142\ \text{K}} \cdot \left(T_{hr}^{\infty} - T_{cw} + \Delta\xi^{(n)}\left(T_{hr}^{\infty}, T_{cw}\right)\right), \quad (6)$$

where T_{ca}^{∞}, T_{hr}^{∞}, and $\Delta\xi$ are, respectively, the cooling air temperature at thermal equilibrium, rotor hot spot temperature at thermal equilibrium, and an nth-order correction factor for the heat exchanger and thermal dynamics not accounted for in the reduced model, which is expressed as a second-order coefficient in Equation 7. These coefficients are obtained by minimizing the maximum absolute error between the quadratic and the H-LPTN models.

$$c = -\frac{\Delta P_r}{78.6142\ \text{K}} \cdot \left(\alpha_1 \cdot \left(T_{hr}^{\infty}\right)^2 + \alpha_2 \cdot T_{hr}^{\infty} + \beta_1 \cdot \left(T_{cw}\right)^2 + \beta_2 \cdot T_{cw} + \gamma\right)$$
$$\text{where: } \min_{\{\alpha_1,\alpha_2,\beta_1,\beta_2,\gamma\}} \max \left|Q_{quad} - Q_{LPTN}\right| \quad (7)$$

2.5 Effective thermal endurance

The insulation thermal endurance can be approximated using the 10 C rule of thumb (derived from the Arrhenius equation) (Janting et al., 2019), as shown in Equation 8, for a time-varying temperature with a periodic profile per day, i.e., boosting hours per day. Furthermore, hydrogenerators facing an increasing number of start–stop cycles and load changes may reduce the life of the equipment; it is estimated to add an *equivalent operating time* (η) between 15 h (Nilsson and Sjelvgren, 1997) and 5 h (EPRI, 2017). Therefore, an effective thermal endurance is proposed in Equation 9 that considers the equivalent operating hours of start–stop cycles with annual growth, where, for each year (m), the added η is calculated up to m and then subtracted from the purely thermal endurance. Subsequently, all the additional η values for each m are interpolated

using radial basis function (RBF) interpolation and then smoothed using a Gaussian filter to create a continuous function, as shown in Equation 10.

$$L(T) = 20000 \cdot 2^{\frac{T_{class}-T}{10}} \quad [\text{h}] \qquad L_{cycle}(\Delta t, T) = \frac{1}{\left[\frac{24-\Delta t}{L(T_{op})} + \frac{\Delta t}{L(T)}\right] \cdot 365} \quad [\text{yr}]$$
$$\text{where: } \{T_{class}, T_{op}\} \Rightarrow \text{constant} \tag{8}$$

$$\rho(t) = \rho_0 \cdot (1+r)^t \quad [\text{start} - \text{stop/yr}] \qquad \Psi_{ss}(m) = \sum_{t=0}^{m} \frac{\eta}{8760} \cdot \rho(t) \quad [\text{yr}]$$
$$\Delta L_{\text{eff}}(\Delta t, T) = L_{cycle}(C_m) - \Psi_{ss}(m) \quad [\text{yr}]$$
$$\forall \quad C_m = \left\{(\Delta t, T) \in \mathbb{R}^2 : L_{cycle}(\Delta t, T) = m\right\}, \quad m \in [0, 100] \subset \mathbb{Z}_0^+ \quad [\text{yr}] \tag{9}$$

$$D = \Delta t \times T = [1, 12] \times [130, 155] \subset \mathbb{R}^2 \quad \text{(domain)}$$
$$I: D \to \mathbb{R} \quad \text{(RBF interpolation)} \quad \Leftrightarrow \quad I(\Delta t, T_{target}) = \sum_i \phi\left(\|\mathbf{x}_i - (\Delta t, T_{target})\|\right) w_i$$
$$\text{where} \quad \phi(z) = z^2 \log z \quad \text{(thin plate spline kernel)}$$
$$\text{and} \quad \mathbf{x}_i \text{ are points from } C_m \text{ sets with weights } w_i \text{ determined by RBF}$$
$$L_{\text{eff}}(\Delta t, T) = (G_\sigma * I)(\Delta t, T) = \iint_{\mathbb{R}^2} \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2+v^2}{2\sigma^2}\right) I(\Delta t - u, T - v)\, du\, dv \quad [\text{yr}] \tag{10}$$

2.6 Reactive boosting energy over the expected lifetime

The reactive boosting energy can be calculated for the additional reactive power injected into the grid, considering the insulation system's effective thermal endurance or lifetime expectancy. In this regard, the reactive boosting energy over the expected lifetime can be maximized as shown in Equation 11.

$$\text{maximize} \quad E_{total}(T, \Delta t, L) = \Delta Q_{boost}(P, T) \cdot \Delta t \cdot 365 \cdot L_{\text{eff}}(T, \Delta t). \tag{11}$$

2.7 Reactive power boosting cost formation

Optimal reactive power dispatch is crucial for dealing with heavily stressed power systems by reducing grid losses and improving voltage stability. In economic optimal reactive power dispatch, the reactive power cost of synchronous generators is typically modeled as a lost-opportunity cost. We propose the Q-strategy framework, which mitigates the opportunity cost associated with the extended ability to contribute reactive power without reducing active power injection. This framework introduces additional generator losses and lifetime costs for reactive power. Thus, the reactive power income is defined as the difference between the maximum reactive power boost and the grid-code maximum reactive power allowance, minus the cost of the additional generator losses beyond the grid code and the cost of lifetime reduction, as illustrated in Figure 4 and formulated in Equation 12. The raw copper material price baseline is taken as λ_{cu} = \$6.5/kg. To translate raw commodity prices into realistic utility project expenditures, an industrial accounting framework is applied, where the specialized labor cost of rewinding constitutes 50%

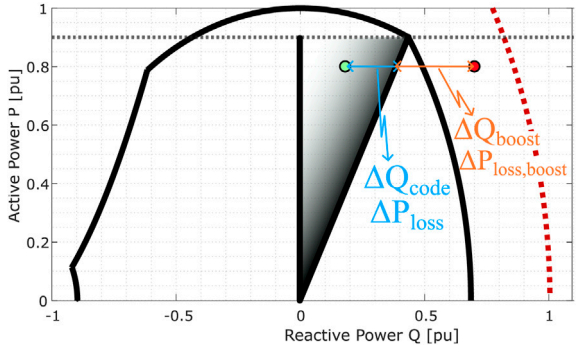


FIGURE 4
Synchronous generator reactive power cost formation outside the grid code. ΔQ_{code} and ΔP_{loss} are considered within the agreement between the plant and grid owners. However, ΔQ_{boost} and $\Delta P_{loss,boost}$ are, respectively, additional reactive power and losses beyond the grid code that must be evaluated.

of the total direct refurbishing expenditure, supplemented by an additional 20% multiplier to account for engineering, insulation materials, and manufacturing overheads. This standard practice yields a total structural multiplier of 2.4 relative to the base copper mass, resulting in an effective fully burdened material replacement cost of \$15.6/kg. Although this effective rate is slightly higher than the standalone raw material baselines evaluated in the recent hydrogenerator refurbishment literature, such as Karekezi et al. (2025), the explicit integration of labor and overhead metrics ensures a more conservative and comprehensive asset-degradation pricing mechanism that protects the plant operator during extreme dynamic boosting regimes.

$$\begin{aligned}\Pi_Q(P, \Delta t, T) &= \lambda_q \cdot \Delta Q_{boost}(P, T) - \lambda_p \cdot \Delta P_{loss,boost}(P) - \lambda_{cu} \cdot m_{cu}/L_{\text{eff}}(\Delta t, T) \\ &= \lambda_q \cdot [Q_{boost}(P, T) - Q_{code}(P)] - \dots \\ &\cdots - \lambda_p \cdot [P_{tot,loss}(P, Q_{boost}(P, T)) - P_{tot,loss}(P, Q_{code}(P))] - \dots \\ &\cdots - 2.4 \cdot \lambda_{cu} \cdot m_{cu}/L_{\text{eff}}(\Delta t, T)\end{aligned} \tag{12}$$

2.7.1 Multi-temperature Pareto front and Chebyshev trade-off analysis

A bi-optimization problem is introduced to analyze the trade-offs between lifetime and income across different operating temperatures. The feasible solution space is defined as in Equation 13:

$$\mathcal{F} = \{(P, \Delta t, T) \mid P \in [0.2, 0.9], \Delta t \in [0, 12], T \in [130\ ^\circ\text{C}, 155\ ^\circ\text{C}]\}. \tag{13}$$

The minimum active power is set to 0.2 pu as a rule of thumb to prevent cavitation. For each temperature T, a bi-objective optimization is solved as in Equation 14:

$$\max_{P, \Delta t} \begin{bmatrix} L_{\text{eff}}(\Delta t, T) \\ \Pi_Q(P, \Delta t, T) \end{bmatrix} \quad \begin{array}{l}\text{where } \Pi_Q(P, \Delta t, T) \text{ is the total income over the lifetime:} \\ \Pi_Q(P, \Delta t, T) = 365 \cdot \Delta t \cdot L_{\text{eff}}(\Delta t, T) \cdot \Pi_q(P, \Delta t, T)\end{array} \tag{14}$$

Since the effective thermal endurance is inversely proportional to temperature and the reactive power income is relatively proportional to temperature, there exists a trade-off between optimizing lifetime and total income over the lifetime. Therefore, a multi-temperature Pareto front analysis is proposed, where a solution $\mathbf{x}^* = (P^*, \Delta t^*)$ is Pareto-optimal if there exists no other feasible solution $\mathbf{x} = (P, \Delta t)$ such that in Equation 15:

$$\begin{cases} L_{\text{eff}}(\Delta t, T) \geq L_{\text{eff}}(\Delta t^*, T) \\ \Pi_Q(P, \Delta t, T) \geq \Pi_Q(P^*, \Delta t^*, T) \end{cases} \quad \text{with at least one strict inequality.} \tag{15}$$

The temperature-specific Pareto front is defined as in Equation 16:

$$\mathcal{P}(T) = \left\{ \left(L_{\text{eff}}(\Delta t, T), \Pi_Q(P, \Delta t, T) \right) \mid (P^*, \Delta t^*) \text{ is Pareto} - \text{optimal at } T \right\}. \tag{16}$$

To find the local trade-off for each temperature T, the first local Utopia points are obtained using Equation 17; then, the Chebyshev distance is applied to find the local trade-offs, as shown in Equation 18, where the trade-off solution for each temperature T is defined as the minimum Chebyshev distance, as shown in Equation 19.

$$U_{local}(T) = \begin{bmatrix} \max\limits_{P,\Delta t} L_{\text{eff}} \\ \max\limits_{P,\Delta t} \Pi_Q \end{bmatrix} \quad \text{for each} \quad T, \tag{17}$$

$$d_C(\mathbf{x}^*, U_{local}(T)) = \max\left\{ \left| 1 - L_{norm}(\mathbf{x}^*) \right|, \left| 1 - \Pi_{norm}(\mathbf{x}^*) \right| \right\} \quad \text{where:} \quad \left\{ L_{norm}(x) = \frac{L_{\text{eff}}(x)}{L_{\text{eff}}^{max}}, \Pi_{norm}(x) = \frac{\Pi_Q(x)}{\Pi_Q^{max}} \right\}, \tag{18}$$

$$\mathbf{X}_{local} = \arg\min_{\mathbf{x}^*} d_C(\mathbf{x}^*, U_{local}) \quad \text{where:} \quad \mathbf{X}_{local} = \{\mathbf{P}_{local}, \Delta \mathbf{t}_{local}\}. \tag{19}$$

The global Pareto front across all temperatures can be represented as in Equation 20. The overall trade-off can be determined using the global Utopia point by minimizing the Chebyshev distance to the global Utopia point, as shown in Equation 21.

$$\mathcal{P}_{global} = \left\{ \left(L_{\text{eff}}(\Delta t, T), \Pi_Q(P, \Delta t, T) \right) \mid \mathbf{y}^* = (P^*, \Delta t^*, T^*) \text{ is globally Pareto} - \text{optimal} \right\}, \tag{20}$$

$$U_{global} = \begin{bmatrix} \max\limits_{P,\Delta t,T} L_{\text{eff}} \\ \max\limits_{P,\Delta t,T} \Pi_Q \end{bmatrix} \quad \mathbf{Y}_{global} = \arg\min_{\mathbf{y}^*} d_C\left(\mathbf{y}^*, U_{global}\right) \quad \text{where:} \quad \mathbf{Y}_{global} = \left\{ \mathbf{P}_{global}, \Delta \mathbf{t}_{global}, \mathbf{T}_{global} \right\}. \tag{21}$$

Finally, an income analysis is performed, where the active power and temperature are varied for each hour trade-off solution as in Equation 22:

$$\Pi_Q(P, \Delta \mathbf{t}_{local}, T) \quad \text{where:} \quad \begin{cases} P \in [0.2, 0.9] \\ T \in [130\ ^\circ\text{C}, 155\ ^\circ\text{C}] \end{cases}. \tag{22}$$

2.8 Hydrogenerator data

We consider four different hydrogenerators for the calculations. The base case for the FEA calculation is G1, while only the scaled modeling calculations are presented for the remaining hydrogenerators. Similarly, the saturation model is based on G1, and the I_f^{ref} values are scaled according to the nominal operating condition.

3 Method: Q-strategy

The Q-strategy is a comprehensive framework that integrates the hydrogenerator's available reactive power capacity, hot spot management for online control, and sufficient incentives for the power producer to offer additional reactive power services to enhance voltage support and grid optimization. As visually formalized via the comprehensive operational flowchart in Figure 5, this framework comprises four sequential pillars for execution: Q-boost, Q-model, Q-energy, and Q-value.

The *Q-boost* pillar evaluates the physical capability of the hydrogenerators to provide extended reactive power compensation beyond standard limits, mapping the over-excited active/reactive operational envelopes $Q(I_a, I_f = \sqrt{T_{max}})$ using a classical machine parameter approach.

These boundaries serve as inputs to the *Q-Model* pillar, which uses the discrete-time CTM of the hydrogenerator to provide accurate, real-time predictions of hot spot temperature. By iteratively processing internal sub-component losses ($\Delta P_f, \Delta P_s, \Delta P_{ex}, \Delta P_{br}, \Delta P_{fe}, \Delta P_v$), the CTM continuously tracks the maximum field current I_f^{max} across various dynamic time windows (Δt = 15 to 650 min) while ensuring that the hotspot never violates the critical continuous thermal threshold of 155 °C.

Next, the *Q-energy* pillar integrates this thermal profile over an asset-management horizon (up to 100 years). By calculating the cyclic start–stop count ρ and equivalent operating hours Ψ_{ss}, it applies a Gaussian filter smoothing technique to determine the effective insulation lifetime reduction L_{eff}, quantifying the net allowable reactive energy injection (E_{total}).

Finally, the *Q-value* pillar establishes the service cost formation engine. It evaluates the marginal additions in machine losses (ΔP_{loss}) and the fully burdened asset degradation cost ($2.4 \cdot \lambda_{cu} \cdot m_{cu}/L_{\text{eff}}$) relative to the baseline grid code requirements (Nøland et al., 2020). By evaluating the boundary condition where the service value function settles to zero ($\mathbb{V}_Q = 0$), it identifies the minimum acceptable market clearing price λ_q.

By seamlessly transitioning from real-time physical constraints to macroeconomic market indices, the overall framework provides transmission system operators (TSOs) with verifiable, thermally secure reactive power-boosting services to enhance grid support.

3.1 Q-boost element

The Q-boost element specifies the reactive power capabilities of hydrogenerators based on thermal limits using a classical POD approach, e.g., the machine designer may provide Q-boost capability. For instance, using Equation 1, the classical Q-boost limits for hydrogenerators at different temperature thresholds are shown in Table 2. It is essential to mention that the ambient and cooling system conditions constrain the maximum temperature, which is assumed to be notwithstanding in this formulation. The IEEE C50.12-2005 states that salient-pole, totally enclosed, water-to-air-cooled hydrogenerators are generally designed for the maximum temperature of the cooling water encountered where the machine is to be installed C50 (2006). Using the proposed H-LTPN for the classical POD approach, the field heating limit for cooling water considerations is between 30 °C and 35 °C, as shown in Figure 6.

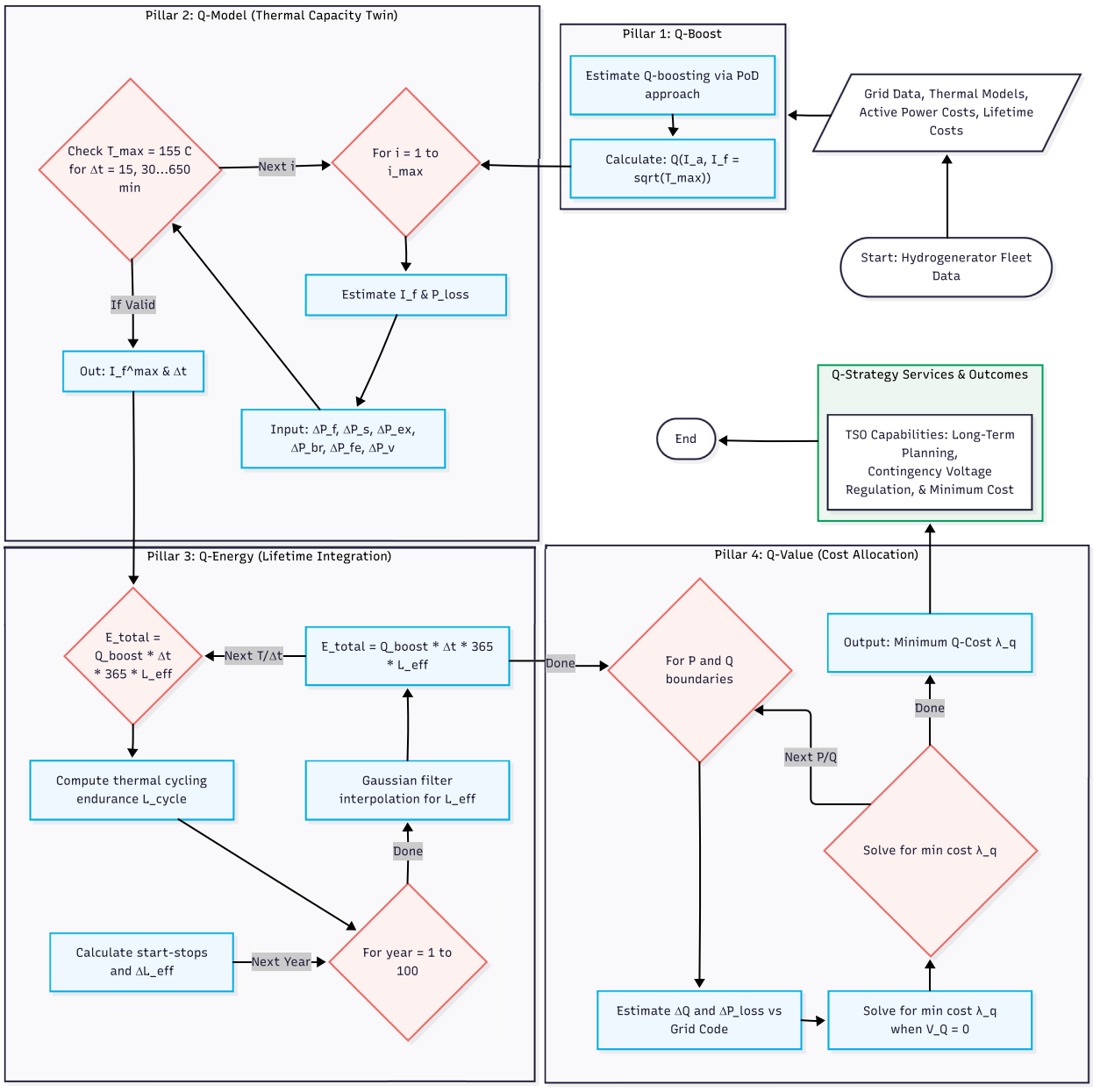


FIGURE 5
Reactive power boosting strategy for the hydrogenerator. The strategy consists of four models: Q-boost, a rule of thumb for determining extended reactive power regions; Q-model, a hot spot model for online control; Q-energy, which provides the maximum reactive power energy over the lifetime of the insulation; and Q-value, which allocates the reactive power boosting cost of the hydrogenerator.

Moreover, the limits are depicted for the temperature thresholds of 155 °C and 180 °C, where the reactive power is enhanced.

Using the proposed H-LPTN, a temperature-dependent capability diagram can be constructed for extended reactive power capabilities according to the rotor hot spot temperature limits, as shown in Figure 7. The difference in reactive power capability between the lowest and highest temperatures, i.e., summer and winter, is presented in Figure 8. The time window consists of $t =$ 12 h, i.e., thermal equilibrium. Thus, considering seasonal changes in cooling water temperature, the hydrogenerator may operate at different thermal limits to provide reactive power services. For instance, during winter conditions, when load increases due to heating, hydrogenerators may deliver additional reactive power to enhance grid support and optimization.

Furthermore, a comparison between the quadratic and H-LPTN models is performed, with a reactive power difference of 2.82% for the relationship between the rotor hot spot and cold water temperatures in G1. The correction factor is $\Delta\xi = [-1.7049 \times 10^{-4},\ 0.9921, -1.4615 \times 10^{-4}, -0.8830, -16.3845\ °C]$. The quadratic model enables ultra-fast calculations of reactive power limits at different seasonal cooling water temperatures, providing a holistic view of reactive power capabilities. For practical purposes,

TABLE 2 Classical approach for estimating reactive power limits at different temperature thresholds.

		T_{class} = 130 °C		T_{class} = 155 °C		T_{class} = 180 °C	
MVA	Gen	I_f^{op} [pu]	Q_{op} [pu]	I_f^{max} [pu]	Q_{max} [pu]	I_f^{max} [pu]	Q_{max} [pu]
103	G1	1.0000	0.4364	1.0919	0.5939	1.1767	0.7333
60	G2		0.5100		0.6694		0.8121
120	G3		0.5103		0.6431		0.7591
12	G4		0.5103		0.6597		0.7928

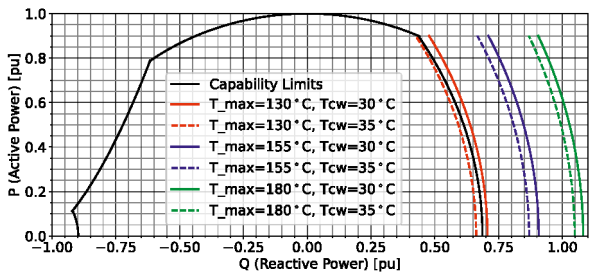


FIGURE 6
Field heating design limits at various temperatures using the proposed H-LTPN.

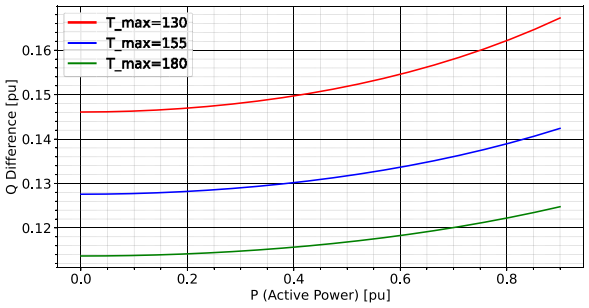


FIGURE 8
Reactive power boosting difference between the lower and upper bounds for the temperature of cooling water of the heat exchanger.

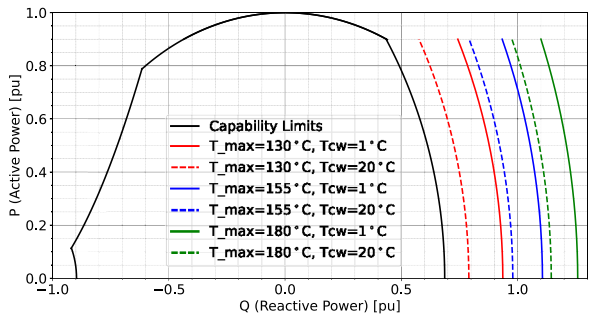


FIGURE 7
Hot spot-thermal capability diagram at the thermal equilibrium reached after 12 h.

the maximum reactive power, at the maximum active power and maximum temperature of the seasonal cooling water, can be taken as the reactive power boosting limit, as presented in Table 3. The Q-boost approximation limits for different real-world hydrogenerators display that the largest machines possess the lowest continuous boosting per unit capabilities. A lower per-unit value of reactive power may be attributed to earlier saturation of the iron core due to the reduced equivalent core length resulting from radial ventilating ducts (Pyrhönen et al., 2013), which are typically present in large hydrogenerators to dissipate heat (Lipo, 2017; Su et al., 2021; Franc et al., 2019).

3.2 Q-model element

The Q-model element proposes using a capacity twin model for real-time operation under steady-state and dynamic thermal conditions for voltage regulation. We proposed two operation modes: restorative and emergency boosting.

1. Restorative boosting for secondary voltage regulation, grid resilience, and optimization, which takes advantage of the thermal margin in the machine. This can be defined as a static Q-boost operation (SQO), in which the thermal limit of the machine is determined for maximum active and reactive power during continuous operation, as shown in Figure 9a. The hydrogenerator may operate continuously at the thermal equilibrium within the temperature limit of the insulation class.
2. **Emergency boosting** for voltage stability support going beyond the reactive power for continuous operation. This necessitates a dynamic operating time window based on the machine's estimated temperature using a CTM, referred to as dynamic Q-boost operation (DQO), as shown in Figure 9b. In this case, the overexcitation and stator current limiters may be disabled so that a thermally controlled voltage regulation can be applied using the CTM.

These two schemes for reactive power boosting are presented in Figure 9 for insulation Class F at P = 0.9 pu, where the static Q-boost refers to a service for which the insulation rating

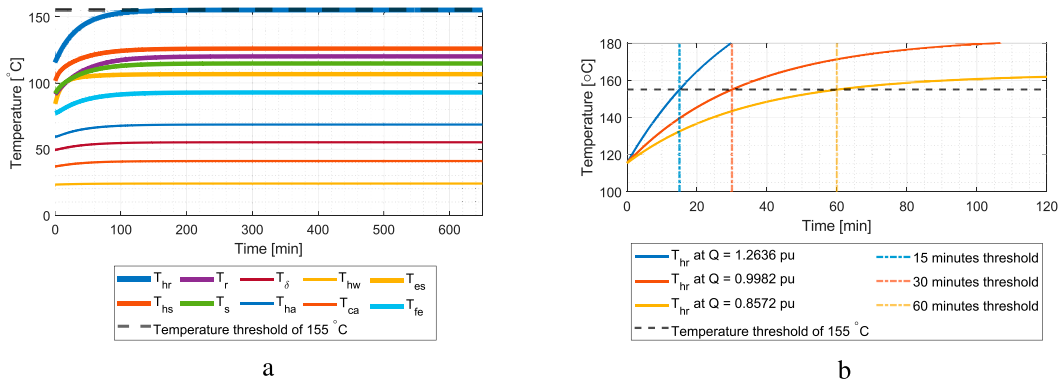


FIGURE 9
Temperature behavior under the two proposed operating modes for the Q-model element for G1. The proposed H-LPTN is used for temperature estimation. **(a)** Static Q-boost operation (SQO) and **(b)** Dynamic Q-boost operation (DQO).

TABLE 3 Static Q-boost approximation using Equation 6 at $T_{hr}^{max} = 155$ °C, $T_{cw} = 20$ °C, and $P = 0.9$ pu.

Gen	MVA	I_f^{max} [pu]	Q_{max} [pu]	Q_{max} [Mvar]
G1	103	1.2217	0.8053	82.9459
G2	60	1.2196	0.8830	52.9800
G3	120	1.2195	0.8154	97.8480
G4	12	1.2274	0.8706	10.4472

class F is utilized without advanced control. In contrast, dynamic Q-boost is the extension of reactive power capability under extreme conditions over different operating time windows, requiring a thermal capacity twin. The simulation uses the following initial temperatures at thermal equilibrium: $T_{cw} = 20$ °C, $T_r = 91.36$ °C, $T_{hr} = 115.37$ °C, $T_s = 92.21$ °C, $T_{hs} = 101.63$ °C, $T_{es} = 84.24$ °C, $T_{fe} = 77.06$ °C, $T_\delta = 49.51$ °C, $T_{ha} = 59.30$ °C, $T_{ca} = 36.99$ °C, and $T_{hw} = 23.20$ °C.

For DQO, especially during the shortest thermal time window of 15 min, the rotor hot spot remains the limiting component for reactive power boosting; the rate of temperature change remains below the recommended temperature gradient of 2.5 °C/min (Istad et al., 2011). Hence, the DQO can be reliably defined for thermal time windows of 15 min or longer, depending on the requirements of the TSO/DSO or the services planned by plant owners.

3.3 Q-energy element

The Q-energy element incorporates the thermal endurance of the hydrogenerator's insulation system into calculating reactive energy dispatched into the grid. This calculation enables Q-boost optimization in grid planning and ORPD, ensuring that Q-boost operation is well aligned with the wear-and-tear-associated costs, i.e., the lifetime reduction cost of the insulation system. Q-energy can be expressed as the product of the Q-boost injected into the grid and the corresponding Q-time interval ($E_Q = Q_{boost} \cdot \Delta t$), which can be used to characterize the effective thermal endurance of the insulation system.

A case example is the maximization of the total reactive energy delivered over the lifetime of a hydrogenerator, as expressed in Equation 11. First, the hydrogenerator's effective thermal endurance is estimated using $\rho_0 = 300$ start–stop cycles (Signore, 2017), with an annual growth $r = 2.8079\%$ (Neupane et al., 2021), considering the added equivalent operating time to be $\eta = 10$ h, and using the on-site measurements of operating points for G1 (Karekezi et al., 2022), the nominal temperature of the rotor hotspots is estimated to be $T_{op} = 80$ °C at $T_{cw} = 20$ °C.

Figure 10 depicts the insulation's effective thermal endurance. It shows the impact of Q-boost at different temperatures in reducing the effective thermal endurance concerning the daily boosting hours—referred to as Q-time. This figure shows that the rotor hot spot temperature must be as close as possible to 130 °C to maximize Q-time, which is directly proportional to the reactive energy, because the reactive power variance is 36%, and the Q-time variance is 1,100%. It should be noted that the Q-time governs the reactive power, estimated based on the hotspot temperature allowance, for reactive energy variation. Therefore, the optimal point for reactive boosting energy over the insulation lifetime is at 130 °C, with 12 hours of daily boosting, yielding 5.4 Tvarh over an 18-year lifespan, while active power is reduced to 0.2 pu. In other words, the minimum temperature with the maximum daily boosting hours is needed to maximize lifetime, and the minimum active power is needed to maximize reactive power boosting.

3.4 Q-value element

The Q-value element develops a reactive power cost formation mechanism that considers hydrogenerator losses, costs beyond the grid requirements, and insulation lifetime costs to incentivize reactive power services. The minimum cost for reactive power (λ_q) can be obtained for an active power cost of $\lambda_p =$ \$70 Mwh and a copper cost $\lambda_{cu} =$ \$6.5kg^{-1} when $\Pi_Q = 0$, and $Q_{max}(P_{max} = 0.9\,\text{pu}, T_{max} = 155\,°\text{C})$. The results for G1, G2, G3, and G4 using Equation 12 are $\lambda_q^{G1} =$ \$0.6872 Mvarh, $\lambda_q^{G2} =$ \$1.0307 Mvarh, $\lambda_q^{G3} =$ \$0.9501 Mvarh, and $\lambda_q^{G4} =$ \$1.3302 Mvarh,

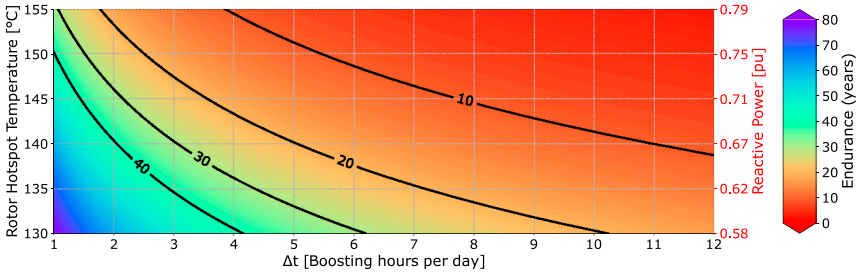


FIGURE 10
Insulation thermal endurance for class F (155 °C).

respectively. Here, the smallest hydrogenerator may require a higher incentive for reactive power-boosting services. Therefore, distributed synchronous generators may require higher reactive power values to incentivize the Q-boost services. However, this reactive power cost may require a holistic approach to the distribution grid as a super-node to provide coordinated reactive power services (Melfald and Øyvang, 2023).

In Figure 11, the reactive power income [$/h] is presented for λ_q^{G1}, in which each subplot refers to a different active power (P). Each subplot depicts different bounds for reactive power boosting because the rotor hot spot temperature limits them. Furthermore, effective endurance is shown for different hours of boosting per day. The least reactive power boosting income occurs at high temperatures, long boosting hours per day, and high active power (losses). In contrast, the most significant income comes from the lowest boosting hours, low temperature, and reduced active power (losses).

The bi-objective optimization analysis of total reactive energy and the lifetime at various temperatures are shown in Figure 12, where the optimal solution is 1700 k$ at 130 °C and 0.2 pu (the lowest active power threshold); the overall Chebyshev trade-off point is 1160 k$ at 130 °C and 0.2 pu, with a lifetime expectancy of 55 years; and the Chebyshev trade-off point for 155 °C is 299 k$ at 0.2 pu, with a lifetime expectancy of 29 years. The multi-temperature Pareto front is performed to illustrate the total income for optimal, utopia, and Chebyshev trade-off points, as shown in Figure 12a; the relationship of the Chebyshev trade-off solutions for temperature and hour considering lifetime is depicted in Figure 12b; and the total income for the Chebyshev trade-off points by variation of active power is presented in Figure 12c. Note that the total income includes the lifetime cost expenses. For instance, the lowest trade-off is presented at 155 °C and a nominal active power of 0.9 pu, where the lifetime expectancy is 28.7 years and the total income is 95 k$. Moreover, for nominal active power operation at 130 °C, the lifetime expectancy increases to 54.8 years, with a total income of 191 k$. These lowest trade-off points have a 67% income variance for a range of 28–55 years of lifetime expectancy. On the other hand, for the trade-off points at the lowest active power, the income variance is 118% over the same lifetime period.

As shown in Figures 11, 12, the optimized Q-value will tend to reduce the operating temperature to achieve the highest daily boosting hours. At the same time, the maximum Q-value is obtained at the lowest possible active power operation because it provides a broader temperature allowance for reactive power boosting. However, the extent of reactive power boosting is ultimately determined by the grid and TSO/DSO requirements.

It is essential to mention that two security measures to protect the hydrogenerators may be present in this approach: 1) the water inlet temperature of the heat exchanger is set to T_{cw} = 20°C and 2) the initial condition or the pre-boost point of operation corresponds to the rated values of the hydrogenerators and, therefore, represents the highest temperature under normal operating conditions.

4 Model validation

The model validation assesses the suitability of different models using real-world data. In this regard, three paths are followed: (1) the H-LPTN cross-validation demonstrates the accuracy and computational efficiency of hot spot temperature predictions. (2) The thermal scaling rule demonstrates the validity of parameter scaling for temperature predictions of four hydrogenerators of different sizes. (3) Two case studies for online thermal control evaluate the adequacy of reactive power boosting for enhanced voltage regulation with temperature control.

Crucially, it must be emphasized that the high-fidelity thermal network validated in this section directly and exclusively governs the *Q-model* pillar—which constitutes one-fourth of the integrated *Q-strategy* framework architecture. Although the remaining operational, lifetime, and economic loops are processed by the *Q-boost*, *Q-energy*, and *Q-value* pillars, respectively, the thermal tracking engine within the *Q-model* serves as the indispensable physical foundation of the entire strategy. It acts as the machine's guardian of structural integrity, ensuring that aggressive grid-side voltage optimization loops never drive the stator or rotor windings into unmonitored localized thermal runaway or catastrophic insulation degradation.

4.1 Simulation environment and machine configuration

To rigorously evaluate the transient capabilities of the proposed Q-strategy, system-level validations are implemented within a single-machine two-bus (SMIB) power system configuration. This simplified electrical topology is intentionally selected to isolate the localized dynamic thermal responses, insulation safety margins, and direct voltage-stability sensitivity of the hydrogenerator, preventing

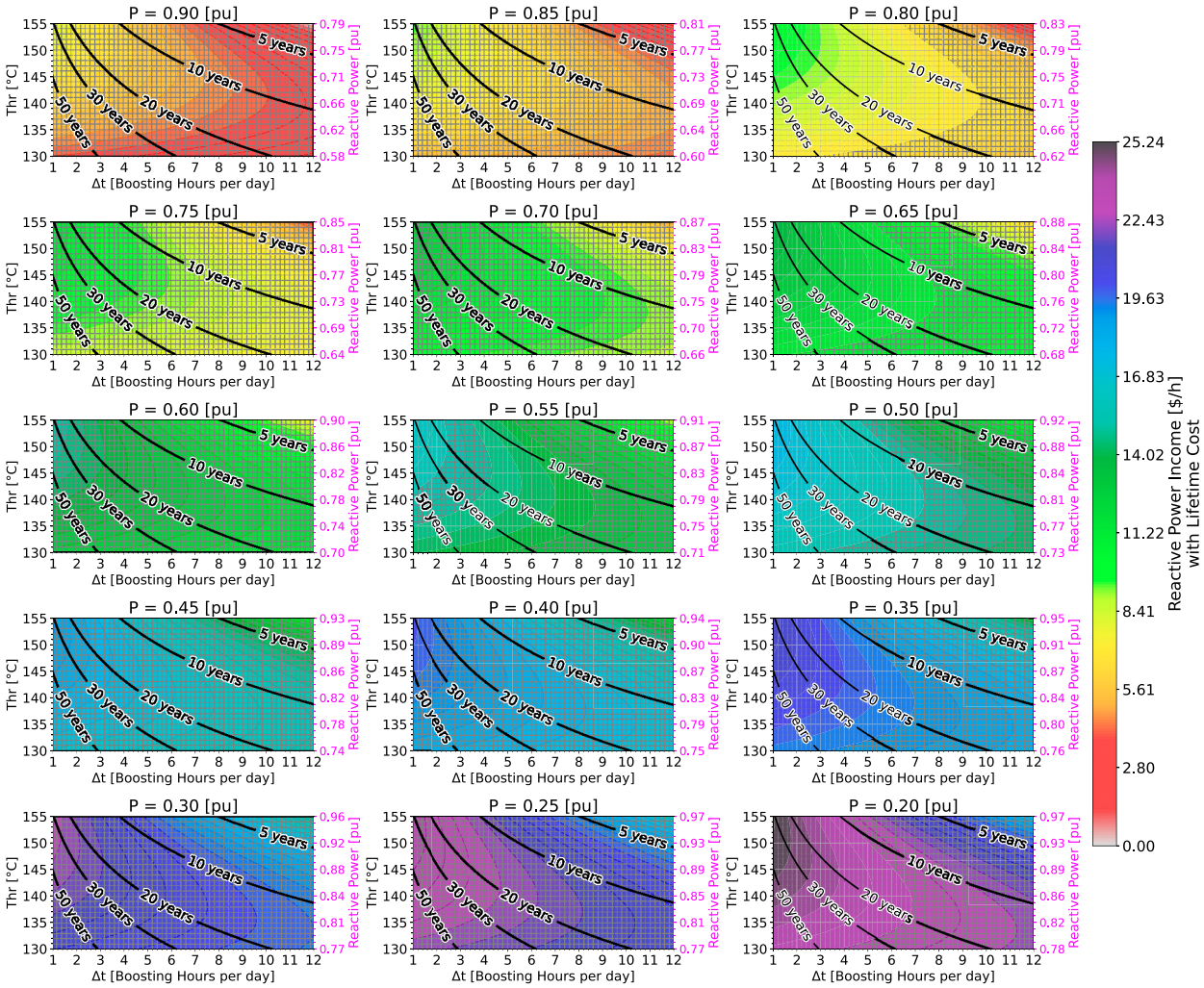


FIGURE 11
Q-value. Reactive power income at different operating points for continuous operation of the hydrogenerator. Here, the reactive power cost λ_q is chosen.

wide-area network impedances or multi-machine control actions from masking the core physical phenomena. Crucially, although the network model is simplified, the context is inherently grounded in real-world Nordic systems; all structural machine parameters, slow open-circuit thermal time constants, and historical start–stop cycle profiles are directly extracted from operational Nordic hydropower statistics. This ensures that the evaluation reflects the exact physical capabilities of the regional asset base under extreme localized stress.

4.2 Hot spot lumped parameter thermal network validation

To ensure the practical reliability of the *Q-Model* safety loops, the underlying H-LPTN was subjected to a rigorous parameter calibration and validation process. The network is calibrated using an optimization pipeline combining a heuristic particle swarm optimization initialization with a formal Bayesian inference framework via Markov chain Monte Carlo (MCMC) sampling. The calibration data comprise empirical-transient and steady-state thermal measurements extracted from official industrial factory shop-floor heat-run tests (HRTs) conducted in compliance with IEC 60034-1, supplemented by localized spatial loss distributions calculated using high-fidelity 2D and 3D electromagnetic FEA solvers.

The resulting low-order model achieves an ultra-fast computational execution speed of 5.34 μs per time-step evaluation, rendering it highly compatible with real-time, multi-machine grid control environments. To establish its self-contained validity across the wide operational envelopes demanded by the *Q-strategy*, the key fidelity metrics and operating conditions are summarized as follows:

- Ultra-high-loss regimes (275% to 508% overloading): Under extreme contingency-level field-current boosting conditions, the maximum prediction error for the limiting structural hot spot temperature remains strictly below 6%.

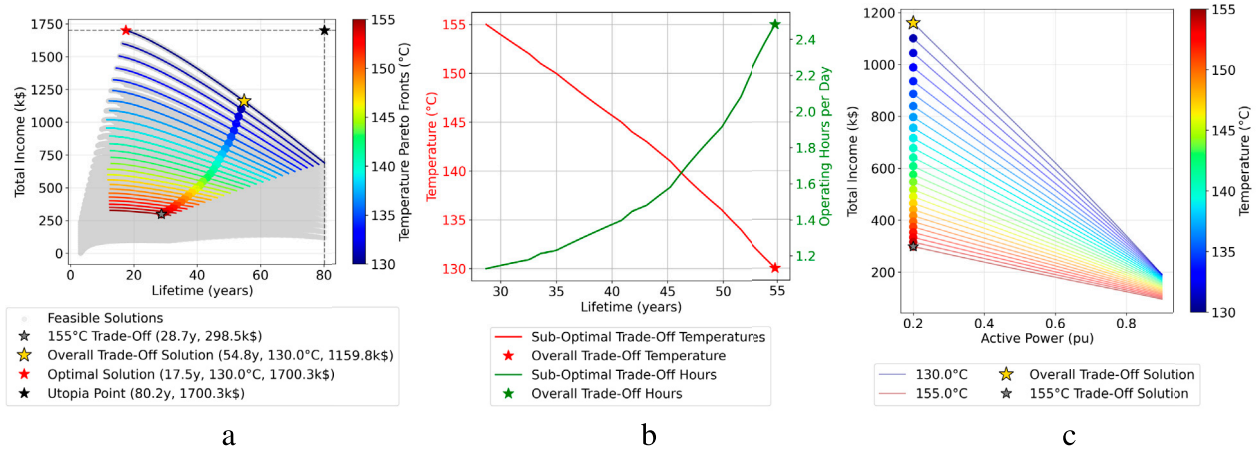


FIGURE 12
Analysis of trade-offs between lifetime, income, temperature, and operating hours for G1: **(a)** Multi-temperature Pareto front with Chebyshev trade-offs when the active power is 0.2 pu; **(b)** Chebyshev temperature and daily boosting hour trade-offs when the active power is 0.2 pu; **(c)** Chebyshev trade-offs extended for different active power.

- High-loss regimes (120% to 201% overloading): Under continuous operational overloading, the thermal network tracks internal temperatures, with an error margin below 9%.
- Transient uncertainty limits: Cross-validation conducted against FEA-derived inverse time characteristic curves demonstrates that the model tracks the limiting thermal components accurately across both fast sub-5-min contingencies and slow, long-term steady-state thermal dynamics (> 5 min).

Although the complete multi-variable parameter estimation algorithm, detailed probability density distributions, and the raw experimental datasets are comprehensive and compiled within the foundational companion study by Tome-Robles et al. (2025), these validated error boundaries (≤ 9 %) directly justify the conservative temperature safety buffers applied to the online temperature limiter (OTL) schemes implemented herein.

4.3 Thermal scaling modeling

The model validation for thermal parameter scaling is performed for three different hydrogenerators. The estimated parameters are presented in Table 4, where G1 is the base case for the scaled thermal parameters of G2, G3, and G4. For instance, the highest percentage difference for the rotor winding hot pot is observed for G4, as expected due to its small size, where the thermal resistance and time constant percentage differences of 394.86% and 6.00%, respectively. Note that the thermal time constant slightly changes due to the intrinsic slow thermal dynamics. In contrast, the thermal resistance exhibits the greatest variation because it determines the temperature variation with respect to losses and the steady-state operating temperature.

The validation for G2 and G3 is performed using heat-run test data obtained during commissioning. The temperature results of the thermal scaling for G2 and G3 are shown in Table 5 and 6, where the limiting component—the rotor temperature—shows differences of up to 4.4% and 12.2% for G2 and G3, respectively. Finally, for G4, the validation is achieved with 24 h real-world temperature measurements. The maximum percentage error over the 24 h comparison is 6.80% at 18.87 h, the valley after the decrease in active power, as presented in Figure 13. The assumptions for G4 are $\Delta T_a =$ 3.3389 K and $\Delta T_w =$ 3.2 K, the air and water temperature difference, respectively, for the heat exchanger dynamics.

4.4 Case studies for online thermal control

This section introduces an OTL scheme, as shown in Figure 14, where a temperature PI controller serves as an outer loop for the AVR to limit the field voltage when the rotor hot spot temperature reaches 150 °C. A secondary voltage control (SVC) uses the load voltage as a pilot voltage to set the reference voltage of the AVR. The two-bus case example shown in Figure 1 is used for the contingency and extreme scenarios. For these examples, the proposed H-LPTN is used for temperature estimation.

4.4.1 Contingency

This example demonstrates the ability of a hydrogenerator with extended reactive power capability to support the grid through voltage enhancement under the following contingencies:

- The SVC is engaged for $V_L =$ 1.025 pu in the 5–365-min interval.
- There is an increase of load ($\Delta L =$ 25%) in the 35–155-min interval.
- A three-phase short-circuit (3ϕ-SC) occurs at Bus 1 at 65 min.
- A 3ϕ-SC occurs at Bus 2 at 95 min.
- One line is disconnected in the 100–250-min interval.
- A 3ϕ-SC occurs at Bus 2 at 130 min.

Figures 15a,b show the electrical and thermal dynamics over an interval of 480 min, during which different contingencies occurred, with the maximum field voltage limited by the rotor hot spot temperature threshold. Notably, during short-circuit and line

TABLE 4 Thermal parameter scaling—thermal resistances and time constants for G1, G2, G3, and G4.

	G1—base				G2			
Component	Loss [kW]	R [K/MW]	Mass [kg]	τ [min]	Loss [kW]	R [K/MW]	Mass [kg]	τ [min]
Rotor copper	191.66	284.7825	9260	36.9188	139	392.6720	7200	39.5809
Stator copper	276.62	55.1734	6827	7.3347	246	62.0409	5150	6.2217
Iron core	211.92	56.8240	66700	17.0403	150	80.2809	41000	14.7984
Stator bar hotspot	488.84	103.4621	73527	5.4649	396	130.0772	46150	4.3125
Stator end winding	276.62	184.2975	6827	8.4545	246	218.6599	5150	6.2960
Rotor winding hotspot	191.66	410.1753	9260	30.3292	139	565.5698	7200	32.5162
	G3				G4			
Component	Loss [kW]	R [K/MW]	Mass [kg]	τ [min]	Loss [kW]	R [K/MW]	Mass [kg]	τ [min]
Rotor copper	251	217.4558	13000	39.5765	33.68	1409.2800	1759	34.7045
Stator copper	387	39.4369	9700	7.4490	56.9	268.2261	1084	5.6618
Iron core	320	37.6317	88800	15.0240	42.77	281.5558	9567.1	12.1106
Stator bar hotspot	707	71.0369	98500	5.0266	99.67	507.9348	10651.1	3.8865
Stator end winding	387	129.1108	9700	7.9345	56.9	900.3067	1084	5.9828
Rotor winding hotspot	251	313.2040	13000	32.5126	33.69	2029.8011	1759	28.5102

TABLE 5 G2 validation with measurements obtained through the HRT at P = 0.845 pu and Q = 0.51 pu, except for the rotor temperature, measured at the nominal operating point.

G2	Measurement	CTM	% Difference
Stator winding	67.9611 °C	78.7899 °C	14.7581
Iron core	52.0333 °C	63.8706 °C	20.4261
Hot air	36.3500 °C	43.4783 °C	17.8591
Cool air	10.7250 °C	17.4621 °C	47.8027
Hot water	8.2000 °C	8.2992 °C	1.2025
Cold water	1.7000 °C	1.7000 °C	0
Rotor copper	75.4000 °C	72.1593 °C	4.3924
Effectiveness (ε)	0.7395	-	-

TABLE 6 G3 validation with measurements obtained through the HRT at P = 0.8575 pu and Q = 0.4825 pu, except for the rotor temperature, measured at the nominal operating point.

G3	Measurement	CTM	% Difference
Stator winding	76.1778 °C	73.6605 °C	3.3600
Iron core	57.5333 °C	58.8851 °C	2.3223
Hot air	39.4000 °C	46.2677 °C	16.0333
Cool air	7.9500 °C	13.5923 °C	52.3835
Hot water	7.0000 °C	7.2026 °C	2.8530
Cold water	1.8000 °C	1.8000 °C	0
Rotor copper	77.5000 °C	68.5736 °C	12.2218
Effectiveness (ε)	0.8364	-	-

disconnections, the temperature dynamics are minimally affected because their time constants are much larger than those of the electrical dynamics. The rotor temperatures are kept constant, whereas the stator and iron core temperatures vary with current and voltage to meet the active power requirements, thereby affecting the stator and iron core losses. Furthermore, during the worst-case contingency period, i.e., the 100–155-min interval, when the line is disconnected, 3ϕ-SC occurs. The load is increased by 25%, and the load voltage reaches 0.96 pu compared to 0.99 pu at the hydrogenerator's terminal, representing a 3.08% difference. Therefore, the grid is supported with additional reactive power by increasing the terminal voltage of the hydrogenerator. A critical remark is that the field voltage is limited to 2.5 pu, which prevents it from reaching the SVC-specified load voltage of 1.025 pu. Nonetheless, the OLT clamps the voltage reference to the AVR to avoid trespassing the maximum allowed hot spot temperature.

4.4.2 Extreme scenario

An extreme scenario is presented where a typical load profile is modeled as a dynamic load with an overall increase of 25%, and the active power generation profile changes every hour according to 24-h real-world data. The voltage control with thermal dependency allows for continuous optimization of the grid over 22 h. Furthermore, the extreme scenario includes the following events:

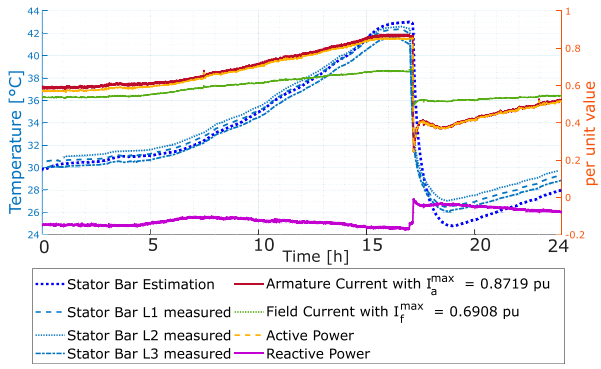


FIGURE 13
Comparison of 24-h real-world measurements with temperature predictions from the G4-scaled CTM . The measurements for the stator coppers L1, L2, and L3 are compared with the stator bar temperature predicted by the CTM, showing good agreement.

- The SVC is engaged for V_L = 1.025 pu in the 5–1,320-min interval.
- A three-phase short-circuit occurs in Bus 1 at 600 min.
- A three-phase short-circuit occurs in Bus 2 at 780 min.
- One line is disconnected in the 785–960-min interval.
- A three-phase short-circuit occurs in Bus 2 at 900 min.

The 24-h extreme scenario shows how reactive power boosting supports the grid under different contingencies for a typical load profile, as shown in Equations 16, 17. The voltage control keeps the rotor hot spot temperature below the threshold and provides sufficient field voltage to prevent voltage collapse during the 785–960-min interval, when a line is disconnected, the peak load is reached, and a 3ϕ-SC is present. Nonetheless, during peaks, the thermal limit is reached, and the voltage at the load no longer follows the set value of 1.025 pu.

In both scenarios, the hydrogenerator provided Q-boost following the SVC pilot voltage while maintaining synchronism during various contingencies and extreme grid conditions.

5 Discussion

The Q-strategy framework enables optimization of existing synchronous generators' reactive power capabilities, potentially avoiding the need for additional FACTS devices for ORPD across multiple time horizons: long-term planning, day-ahead scheduling, reactive power pre-dispatch, and real-time balancing. In this regard, synchronous generators with the capabilities for reactive power boosting (Q-SG) may be used as follows:

- Long-term planning: Allocate the reactive power capability of Q-SGs, with a 9% and 17% increase in field current from the classical operating condition for insulation classes F and H, respectively.
- Day-ahead: Schedule Q-SGs with the quadratic approximation for the static Q-boost limit, providing a straightforward calculation for the maximum reactive power of Q-SG.
- Q-balance: Incorporate an online hot spot temperature-dependent voltage control for reactive power ancillary service market, grid optimization, and contingencies.

The main challenges faced by reactive power boosting are thermal cycling, power-electronics thermal limits, and harmonization with modern grid codes. The Q-strategy implementation may require several technical considerations. A grid-independent excitation system for extended dynamics may be necessary, alongside careful evaluation of adaptive cooling systems. In the case of power electronics, they have a poor thermal constant during extended operation (Rodriguez et al., 2021; Kuprat et al., 2023). Excitation system modifications that incorporate capability curves and a dedicated current-limiter, as proposed by Ojo and Shafiul Alam (2026) for safe synchronous condenser operation of hydrogenerators, illustrate the kind of control upgrades that a comprehensive Q-Strategy demands. Therefore, resizing the hydrogenerator's reactive power-boosting capabilities will require a new design philosophy aligned with these new services. The approach demands a comprehensive digital twin model that encompasses the generator, step-up transformers, excitation systems, and other components, supported by additional sensors and advanced control systems.

Winding refurbishment to a higher thermal threshold should be studied. For instance, using the thermal endurance rule-of-thumb equation, the total lifetime of the insulation class F for 12 h of boosting at 155 °C per day is 4.4511 years. However, if an insulation system upgrade is considered, this can be increased for an insulation class H up to 25.6885 years. In this regard, 21.1474 years are added to the hydrogenerator, with the possibility of delivering reactive power-boosting services to the grid, that is, an increase of 130.45%. Complementary to insulation upgrades, rotor pole redesign and field winding refurbishment can further reduce ohmic losses and shift the optimal reactive power characteristics, with net economic benefits shown to be four to five times the incurred material costs (Karekezi et al., 2025). This new design philosophy can be accompanied by an update of IEEE (2006) to incorporate peak capabilities for insulation classes F and H, similar to the IEEE (2014), where class H insulation can be the next step for rotor winding.

Regarding the grid code harmonization, the modern grid code establishes a static envelope (Q/P_{max} = 0.495) (Nøland et al., 2020). However, if the reactive power output for G1 may be up to Q = 0.79 pu for continuous operation, the maximum active power at the connection point should be P_{max} = 1.6 pu to comply with the grid code. Therefore, making the codes flexible through dynamic operating envelopes is indispensable, especially under restorative and emergency operating conditions. Future work should address fast reactive power-boosting response, reactive power-boosting coordination, and unit commitment. Q-energy can be further developed considering the TSO/DSO grid codes, contingencies, and optimization for multi-machine reactive power compensation, coordination, and synchronization through the ORPD. In multi-machine coordination scenarios (Melfald and Øyvang, 2023; Dragosavac et al., 2012; Dragosavac et al., 2013), the Q-strategy can maximize reactive power allocation benefits by incorporating these dynamic envelopes for grid codes and managing OEL thresholds through online hot spot temperature predictions using a CTM, while considering insulation system lifetime costs.

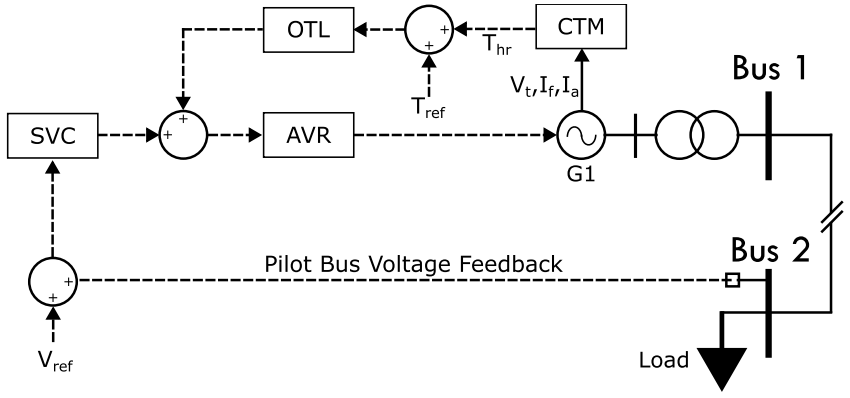


FIGURE 14
Online temperature limiter block diagram. The OTL is a temperature PI controller.

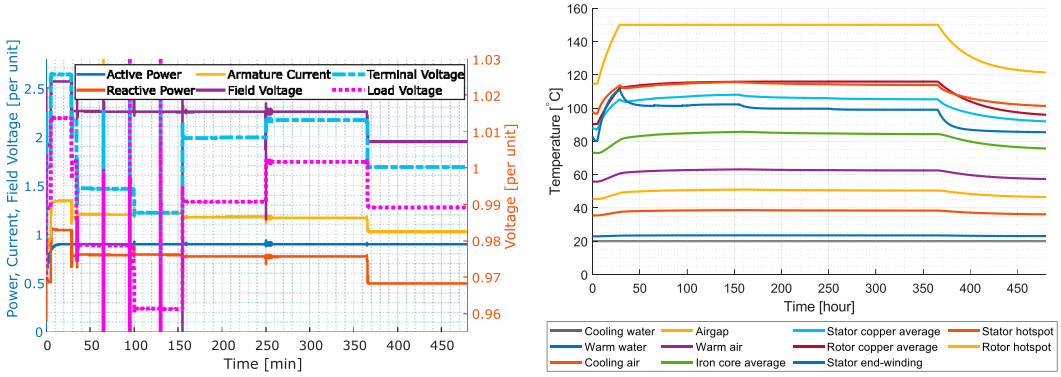


FIGURE 15
System behavior during a series of contingencies: **(a)** electrical response and **(b)** thermal response.

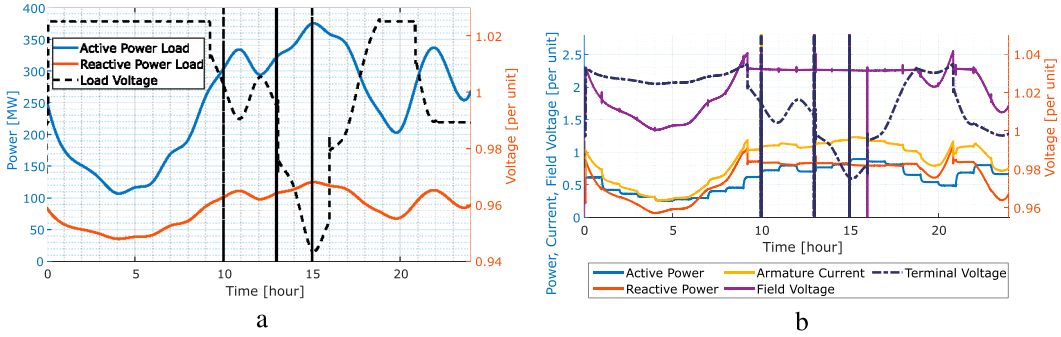


FIGURE 16
System electrical behavior during a 24-h extreme scenario: **(a)** load response and **(b)** generator response.

Furthermore, thermal control is crucial for minimizing wear and tear through temperature change rate management and Q-boost time limitations to comply with the standard for thermal cycling. The daily boosting duration, defined as Q-time (Δt), may be used to minimize the tear and wear of the hydrogenerator while considering the power plant owner's requirements.

Therefore, hot spot temperature management demands rethinking the machine philosophy of design, control schemes, cooling systems, and copper insulation class temperature margins.

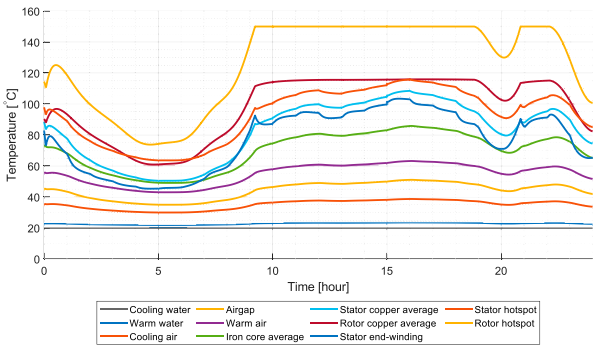


FIGURE 17
Temperature behavior during a 24-h extreme scenario.

In this study, a CTM for air-cooled hydrogenerators is proposed as a significant tool to mitigate the risks and costs associated with thermal dynamics, demonstrating how advanced control systems can accurately predict the operational efficiency, safety, and longevity of hydrogenerators.

This study not only presents a pioneering approach to addressing the challenges posed by the integration of VREs but also lays the foundation for future research to refine and expand the adaptive multi-component thermal hot spot management methodology to enhance the operational efficiency and reliability of hydropower plants.

5.1 Framework boundaries and limitations

Although the quantitative results demonstrate a significant opportunity for utilizing hydrogenerator thermal margins to support voltage stability, several critical framework boundaries and areas for future refinement must be acknowledged to prevent over-generalization:

1. Network scale and generalizability to multi-bus systems: The system-level electrical validation presented in this study is strictly limited to a single-machine, two-bus topology. This simplified framework was intentionally selected to isolate local electro-thermal dynamics and insulation safety margins. Consequently, the findings do not capture wide-area multi-machine transient interactions, spatial voltage profiles across weak transmission corridors, or multi-bus network congestion. Scaling the Q-strategy for realistic regional grid integration requires future compatibility with multi-machine ORPD coordination algorithms, in which the proposed reactive power cost formation can be evaluated within a broader system-wide network optimization context.
2. Dependence on hydrogenerator-specific thermal margins: The boosting boundaries derived herein are strongly dependent on the physical design margins unique to hydrogenerators—namely, the widespread historical implementation of Class F insulation (155 °C) operating under highly conservative Class B (130 °C) operational baselines. Synchronous machines with tighter original design tolerances, alternative cooling systems, or degraded baseline insulation will exhibit significantly narrower boosting windows. The results should not be loosely generalized to general synchronous asset classes without an asset-specific thermal margin audit.
3. Cost model parametrization and assumptions: The economic valuation under the Q-value pillar is inherently sensitive to localized market structures, the volatility of copper price indices (λ_{cu}), and specific national grid code architectures. The parameters applied represent a standardized baseline and should be interpreted as illustrative of an engineering and market setting rather than a universal or globally applicable tariff configuration.
4. Physical infrastructure and experimental requirements: Real-world deployment of the Q-strategy framework demands physical control modifications to existing automatic voltage regulator (AVR) systems and the replacement of conventional static OELs with dynamic, temperature-dependent current-limiting protection blocks. Crucially, the proposed online thermal controller requires deeper validation using hardware-in-the-loop (HIL) testing in real-world environments before full utility deployment.
5. Modeling fidelity and electro-thermal approximations: The underlying physical models operate under specific boundary approximations. The field current models in hyper-saturation regions require further verification to ensure absolute reliability under extreme excitation, and the heat exchanger models must be expanded to establish a standardized thermal-scaling-equivalent procedure valid across all ambient temperatures. Finally, the tracking precision of the hot spot data can be enhanced by substituting or validating parameters with ultra-high-fidelity 3D computational fluid dynamics (CFD) simulations.
6. Control-theoretic and verification bounds: The operational evaluation of the proposed Q-strategy framework—including the OTL and secondary voltage control loops—is strictly grounded in empirical data calibration, multi-scale numerical simulations, and historical operational statistics. The framework has not been formally verified in a strict mathematical or control-theoretic sense. This study does not supply analytical proofs of closed-loop stability, convergence guarantees, or formal logical model-checking certificates for the coupled multi-variable thermal-economic loops. Establishing formal control stability bounds and algebraic safety guarantees for the multi-pillar optimization engine constitutes a vital and promising direction for future mathematical research.

6 Conclusion

The Q-strategy framework provides new services to support and optimize the grid for normal and extreme scenarios, such as restorative or emergency boosting, especially for a high share of VREs. This intelligent management of the untapped potential of hydrogenerators can be scheduled in long-, medium-, or short-term markets following the four elements:

- *Q-boost:* Extends hydrogenerator reactive power output by 57%, considering the actual thermal insulation threshold.
- *Q-model:* Provides a fast thermal model for hot spot online control, with a prediction error up to 8.9%.
- *Q-energy:* Optimizes reactive power boosting over the hydrogenerator's insulation lifespan.
- *Q-value:* Incentivizes hydropower plants via a cost model that considers losses and lifetime depreciation of the machine.

This novel framework enhances hot spot management in the hydrogenerator to optimize their reactive power capacities, keeps temperatures within the insulation class thermal thresholds under contingencies due to its relatively large thermal time constants, and enables reactive power cost formation by considering additional losses and lifetime costs. The H-LPTN can be used as a generic CTM for air-cooled hydrogenerators of different sizes and shows good agreement with temperature measurements and hot spot FEM calculations. Furthermore, the quadratic thermal model is shown to agree well with the H-LPTN for ultra-fast calculations based on rotor losses and field current.

The disruptive step toward implementing the Q-strategy framework comes from the economic incentives of reactive power income, considering additional losses and lifetime cost, where it is found that a trade-off solution to maximize lifetime and income can be in a range of 299–1,700 k$ for 155 °C to 130 °C, respectively, when the active power is reduced to 0.2 pu. That means an income percentage difference of 140%. On the other hand, when the active power is at a nominal value of 0.9 pu, the income range is 95–191 k$ for 155 °C to 130 °C, respectively. The income percentage difference is 67%. Note that this income already includes the lifetime cost for winding refurbishment.

This study finds that smaller generators require higher reactive power costs to incentivize the Q-boost services. Hence, distributed generation might require a separate assessment of reactive power costs. Moreover, the most efficient generator will produce cheaper reactive power when the generators are similar. For a 12-MVA hydrogenerator, the reactive power cost is $\lambda_q =$ 1.3302 Mvarh. In contrast, the most efficient hydrogenerators have the lowest costs, i.e., for a 103-MVA hydrogenerator with a reactive power cost of $\lambda_q =$ 0.6872 Mvarh. The percentage difference between the highest and lowest reactive power costs is 63.75%.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material; further inquiries can be directed to the corresponding author.

Author contributions

DT-R: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review and editing, Data curation, Formal analysis, Software, Validation, Visualization. FM: Conceptualization, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review and editing. JN: Conceptualization, Project administration, Supervision, Validation, Writing – original draft, Writing – review and editing. SM: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review and editing. TØ: Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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Nomenclature

ΔP_f	Field winding losses, [pu] or [MW]
ΔP_r	Rotor total losses, [pu] or [MW]
ΔP_s	Stator winding losses, [pu] or [MW]
ΔP_{br}	Field brush losses, [pu] or [MW]
ΔP_{ew}	Stator end-winding losses, [pu] or [MW]
ΔP_{ex}	Exciter losses, [pu] or [MW]
ΔP_{fe}	Iron core losses, [pu] or [MW]
ΔP_{hr}	Rotor hot spot losses, [pu] or [MW]
ΔP_{hs}	Stator hot spot losses, [pu] or [MW]
ΔP_v	Ventilation losses, [pu] or [MW]
Δ_t	Q-time or reactive power boosting time, [h/day]
λ_e	Conductor's electrical conductivity, [S/m]
λ_p	Active power cost, [$/Mwh]
λ_q^{Gx}	Reactive power cost for the hydrogenerator G_x, [$/Mvarh]
λ_{cu}	Copper cost, [$/kg]
Ψ	Calculated wear and tear for each year, [−]
$\rho(t), \rho_0$	Number of start–stop cycles for each year, [−]
AWT	Accumulated wear and tear, [year]
Π_Q	Lifetime reactive power cost income, [$]
Π_Q	Reactive power cost income, [$/h]
τ_i	Thermal time constant at node i, [m]
τ_{es}	Stator end-winding thermal time constant, [min]
τ_{fe}	Iron core thermal time constant, [min]
τ_{hr}	Rotor hot spot thermal time constant, [min]
τ_{hs}	Stator hot spot thermal time constant, [min]
τ_r	Rotor winding thermal time constant, [min]
τ_s	Stator winding thermal time constant, [min]
ε	Heat's exchanger effectiveness
A_{hr}	Cross-sectional surface area of the conductor, [m^2]
b_f	Width of the pole coil, [m]
E_p	Potier voltage, [pu] or [V]
E_Q	Reactive energy, [Mvarh]
E_{total}	Reactive energy over lifetime [Mvarh]
I_a	Armature winding current, [pu] or [A]
I_f	Saturated field winding current, [pu] or [A]
J_f	Field current density, [pu] or [A/mm^2]
L	Insulation thermal endurance, [h]
L_{cycle}	Thermal cycling endurance, [year]
L_{eff}	Effective thermal endurance, [year]
m	Mass of material, [kg]
P	Active power, [pu] or [MW]
Q	Reactive power, [pu] or [MVAr]
R_δ	Air convection thermal resistance, [K/MW]
R_{es}	Stator end-winding thermal resistance, [K/MW]
R_{fe}	Iron core thermal resistance, [K/MW]
R_{hr}	Rotor hot spot thermal resistance, [K/MW]
R_{hs}	Stator hot spot thermal resistance, [K/MW]
R_r	Rotor winding thermal resistance, [K/MW]
R_s	Stator winding thermal resistance, [K/MW]
R_w	Water convection thermal resistance, [K/MW]
$T_\delta(t)$	Air gap temperature, [℃] or [K]
$T_{ca}(t)$	Cold air heat exchanger temperature, [℃] or [K]
T_{class}	Temperature threshold of insulation, [℃]
$T_{cw}(t)$	Cold water heat exchanger temperature, [℃] or [K]
$T_{es}(t)$	Stator end-winding temperature, [℃] or [K]
$T_{fe}(t)$	Iron core average temperature, [℃] or [K]
$T_{ha}(t)$	Hot air heat-exchanger temperature, [℃] or [K]
$T_{hr}(t)$	Rotor winding hot spot temperature, [℃] or [K]
$T_{hs}(t)$	Stator winding hot spot temperature, [℃] or [K]
$T_{hw}(t)$	Hot water heat-exchanger temperature, [℃] or [K]
$T_r(t)$	Rotor winding average temperature, [℃] or [K]
$T_s(t)$	Stator winding average temperature, [℃] or [K]
V_L	Load voltage, [pu] or [V]
V_t	Terminal voltage, [pu] or [V]
X_{line}	Line reactance, [pu] or (Ω)
r	Annual growth of start–stop cycles, [−].